



Architect Korner File Management

Software architects have a lot to do with reading and writing to databases, files, devices and memory locations. Sending data in and out of files efficiently and synchronously is very crucial in this field. Files are important not only from an operating system perspective, but also for application programming. This article introduces you to how architects use files to build their applications, middleware and entire software products on the Linux platform.

The word 'file' originates from the Latin *filum*, which means a thread to hold loose papers, especially arranged for reference. But in computing science, the word 'file' is used as a collection of data stored with one name, which is the file name. Linux treats everything as a file. As an example, data, devices, inter-process communication mechanisms, and links are all dealt with as

files. There are different types of files—namely, regular or ordinary files, directory files, device files, link files, FIFO and socket files. Device files are of two types: character and block device files. There are two types of link files: soft and hard [1] link files. In Linux, these files are arranged in an inverted tree data structure, for easy maintenance.

How are these files stored on a storage medium like a hard disk, USB storage or